Length of circular arc

1

a 11.5 units

b 19.6 units

2

a 10.47 cm

b 33.25 mm

c 9.42 cm

d 6.00 cm

3

a $\frac{5\pi}{6}$ radians

b $\frac{25\pi}{2} \text{ cm}$

4 $\left(20 + \frac{25\pi}{3}\right) \text{ cm}$

5

a 36.8 cm

b 16.8 cm

6 $\theta = 1.95$ radians

7

a $\theta = 0.9$ radians

b $\theta = 1.1$ radians

8

a $\theta = 1.67$ radians

b $\theta = 96^\circ$

9 $70\pi \text{ cm}$

Area of a sector

10

a 463.10 m^2

b 313.01 m^2

11 $39\pi \text{ units}^2$

12

a 50 cm^2

b 729.87 m^2

c 612.42 m^2

13

a 2π

b $108\pi \text{ cm}^2$

14

a $\theta = 2$ radians

b $\frac{9}{4}\pi^2$ cm^2

15

a 6π cm^2

b 6π cm^2

c 6π cm^2

16

a $\angle JOK = 46^\circ$

b 90 cm^2

17 81 cm^2

18 317.9 m^2

19

a 48π cm^2

b 336π cm^2

20

a 10π cm^2

b $(5\pi + 8)$ cm

21 $\theta = 1.7$ radians

22

a $\theta = 0.8$ radians

b $l = 16$ m

23 $l = 11$ cm

Triangles in sectors

24

a $\angle AOC = 0.76$ radians

b Arc $AB = 44.08$ cm

25

a $AB = 12$ cm

b $(3\sqrt{2}\pi - 12)$ cm

26

a $AB = 7.2$ cm

b $x = 7.0$ cm

c 28.8 cm^2

27

a 49 cm^2

b $\left(\frac{49\pi}{3} - 49\right)$ cm^2

28 $(18\pi - 36\sqrt{2})$ cm^2

29

a

i 61.99 cm^2

ii 383.33 cm^2

b

i 126.90 cm^2

ii 18.90 cm^2

30

a $\angle ROP = 1.02$ radians

b 42.43 cm

31 $(100\sqrt{3} - 50\pi)$ cm^2